Progress Check

1. Euclid's division algorithm is a repeated application of division lemma until we get remainder as _____.
2. The HCF of two equal positive integers k, k is _____.

Illustration 1

Using the above Algorithm, let us find HCF of two given positive integers. Let $a = 273$ and $b = 119$ be the two given positive integers such that $a > b$.

We start dividing 273 by 119 using Euclid's division lemma.

we get, $273 = 119 \times 2 + 35$ …(1)

The remainder is $35 \neq 0$.

Therefore, applying Euclid's Division Algorithm to the divisor 119 and remainder 35. we get,

$119 = 35 \times 3 + 14$ …(2)

The remainder is $14 \neq 0$.

Applying Euclid's Division Algorithm to the divisor 35 and remainder 14.

we get, $35 = 14 \times 2 + 7$ …(3)

The remainder is $7 \neq 0$.

Applying Euclid's Division Algorithm to the divisor 14 and remainder 7.

we get, $14 = 7 \times 2 + 0$ …(4)

The remainder at this stage $= 0$.

The divisor at this stage $= 7$.

Therefore, Highest Common Factor of 273, 119 $= 7$.

Example 2.4 If the Highest Common Factor of 210 and 55 is expressible in the form $55x - 325$, find x.

Solution Using Euclid's Division Algorithm, let us find the HCF of given numbers

$$\begin{aligned} 210 &= 55 \times 3 + 45 \\ 55 &= 45 \times 1 + 10 \\ 45 &= 10 \times 4 + 5 \\ 10 &= 5 \times 2 + 0 \end{aligned}$$

The remainder is zero.

So, the last divisor 5 is the Highest Common Factor (HCF) of 210 and 55.

$\because$ HCF is expressible in the form $55x - 325 = 5$

$$\begin{aligned} \Rightarrow \quad 55x &= 330 \\ x &= 6 \end{aligned}$$

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